MachineLearning

exercise1.py .../WilliamsBenjamin_Lab5/... U | Student-Pass-Fail.csv .../Exercise1 U | exercise2.py .../W

Lab 5 > WilliamsBenjamin_Lab5 > Exercise1 > exercise1.py > ...

```python
import numpy as np
import pandas as pd
from sklearn.linear_model import LogisticRegression

df = pd.read_csv("Student-Pass-Fail.csv")

X = df.iloc[:, 0:2].values
y = df.iloc[:, 2].values

model = LogisticRegression()
model.fit(X, y)

odds = np.exp(model.coef_)
print("Odds:", odds)

points = np.array([[7, 28], [10, 34], [2, 39]])
probs = model.predict_proba(points)

for p, pr in zip(points, probs):
    print("Data point:", p.tolist(), "Probabilities:", pr.tolist())
```

OUTPUT ... Filter Tasks

CHAT TERMINAL

```
  File "/Users/benwilliams/Documents/Development/MachineLearning/.venv/lib/python3.13/site-packages/pandas/io/parsers/readers.py", line 1904, in _make_engine
    self.handles = get_handle(
                   ~~~~~~~~~~^
        f,
        ^^
    ...<6 lines>...
        storage_options=self.options.get("storage_options", None),
        ^^^
```

MachineLearning

EXPLORER

- MACHINELEARNING
  - .venv
  - .vscode
    - settings.json
  - HW
    - HW4
    - WilliamsBenjamin_HW1
      - Exercise1
      - Exercise2
      - Exercise3
      - HW1.pdf
      - WilliamsBenjamin_HW1.zip
  - Lab 1
  - Lab 2
    - WilliamsBenjamin_Lab2
    - Lab2.pdf
    - WilliamsBenjamin_Lab2.zip
  - Lab 5 / WilliamsBenjamin_Lab5
    - Exercise1
      - exercise1.py U
      - image.png U
      - Student-Pass-Fail.csv U
    - Exercise2
      - exercise2.py U
      - Student-Pass-Fail.csv U
    - Exercise3
      - Bank-data.csv U
      - exercise3.py U
  - .gitignore

OUTLINE

TIMELINE

Benjamin_Lab5/... U | Student-Pass-Fail.csv .../Exercise1 U | exercise2.py .../WilliamsBenjamin_Lab5/... U

Lab 5 > WilliamsBenjamin_Lab5 > Exercise2 > exercise2.py > ...

```python
import numpy as np
import pandas as pd
from sklearn.linear_model import LogisticRegression

def split_data(X, y, test_size):
    n = len(X)
    split = int(n * (1 - test_size))
    X_train = X[:split]
    y_train = y[:split]
    X_test = X[split:]
    y_test = y[split:]
    return X_train, X_test, y_train, y_test

def accuracy_score_manual(y_true, y_pred):
    return np.mean(y_true == y_pred)

def confusion_matrix_manual(y_true, y_pred):
    labels = np.unique(np.concatenate([y_true, y_pred]))
    m = np.zeros((len(labels), len(labels)), dtype=int)
    idx = {lab: i for i, lab in enumerate(labels)}
    for t, p in zip(y_true, y_pred):
        m[idx[t], idx[p]] += 1
    return labels, m

df = pd.read_csv("Student-Pass-Fail.csv")
X = df.iloc[:, 0:2].values
y = df.iloc[:, 2].values

test_size = float(input("Enter test size (e.g., 0.2): "))
X_train, X_test, y_train, y_test = split_data(X, y, test_size)

model = LogisticRegression()
model.fit(X_train, y_train)

y_pred = model.predict(X_test)

acc = accuracy_score_manual(y_test, y_pred)
labels, cm = confusion_matrix_manual(y_test, y_pred)

print("Accuracy:", acc)
print("Labels order:", labels.tolist())
print("Confusion Matrix:\n", cm)
```

CHAT | TERMINAL

```
^^^^^^^
    )
    ^
  File "/Users/benwilliams/Documents/Development/MachineLearning/.venv/lib/python3.13/site-packages/pandas/io/common.py", line 926, in get_handle
    handle = open(
        handle,
    ...<3 lines>...
        newline="",
    )
FileNotFoundError: [Errno 2] No such file or directory: 'materialsOutliers.csv'
benwilliams@Bens-MacBook-Air-2 Exercise5 % /Users/benwilliams/Documents/Development/MachineLearning/.venv/bin/python /Users/benwilliams/Documents/Development/MachineLearning/HW/HW4/WilliamsBenjamin_HW4/Exercise5/exercise5.py
Final Coefficients: [-0.26023958  0.05335903  0.18666974]
Intercept: 189.89442410629354
benwilliams@Bens-MacBook-Air-2 Exercise5 % cd ../../..
benwilliams@Bens-MacBook-Air-2 HW % cd ..
benwilliams@Bens-MacBook-Air-2 MachineLearning % cd Lab 5
cd: string not in pwd: Lab
benwilliams@Bens-MacBook-Air-2 MachineLearning % ls
HW      Lab 1   Lab 2   Lab 5
benwilliams@Bens-MacBook-Air-2 MachineLearning % cd Lab\ 5
benwilliams@Bens-MacBook-Air-2 Lab 5 % cd WilliamsBenjamin_Lab5
benwilliams@Bens-MacBook-Air-2 WilliamsBenjamin_Lab5 % ls
Exercise1       Exercise2       Exercise3
benwilliams@Bens-MacBook-Air-2 WilliamsBenjamin_Lab5 % cd Exercise1
benwilliams@Bens-MacBook-Air-2 Exercise1 % /Users/benwilliams/Documents/Development/MachineLearning/.venv/bin/python "/Users/benwilliams/Documents/Development/MachineLearning/Lab 5/WilliamsBenjamin_Lab5/Exercise1/exercise1.py"
Odds: [[12.49500887  0.42784772]]
Data point: [7, 28] Probabilities: [0.24645934369530575, 0.7535406563046942]
Data point: [10, 34] Probabilities: [0.026606156436529083, 0.9733938435634709]
Data point: [2, 39] Probabilities: [0.9999999991172047, 8.827953286298422e-10]
benwilliams@Bens-MacBook-Air-2 Exercise1 % /Users/benwilliams/Documents/Development/MachineLearning/.venv/bin/python "/Users/benwilliams/Documents/Development/MachineLearning/Lab 5/WilliamsBenjamin_Lab5/Exercise1/exercise1.py"
Odds: [[12.49500887  0.42784772]]
Data point: [7, 28] Probabilities: [0.24645934369530575, 0.7535406563046942]
Data point: [10, 34] Probabilities: [0.026606156436529083, 0.9733938435634709]
Data point: [2, 39] Probabilities: [0.9999999991172047, 8.827953286298422e-10]
benwilliams@Bens-MacBook-Air-2 Exercise1 % cd ../Exercise2
benwilliams@Bens-MacBook-Air-2 Exercise2 % /Users/benwilliams/Documents/Development/MachineLearning/.venv/bin/python "/Users/benwilliams/Documents/Development/MachineLearning/Lab 5/WilliamsBenjamin_Lab5/Exercise2/exercise2.py"
Enter test size (e.g., 0.2): .2
Accuracy: 0.975
Labels order: [0, 1]
Confusion Matrix:
 [[ 91   5]
 [  0 104]]
benwilliams@Bens-MacBook-Air-2 Exercise2 %
```

MachineLearning

Student-Pass-Fail.csv .../Exercise2 U | .gitignore | exercise3.py .../WilliamsBenjamin_Lab5/... U

Lab 5 > WilliamsBenjamin_Lab5 > Exercise3 > exercise3.py > ...

```python
import numpy as np
import pandas as pd
from sklearn.linear_model import LogisticRegression

df = pd.read_csv("Bank-data.csv")

df = df.iloc[:, 1:]
X = df.iloc[:, :-1].values
y = df.iloc[:, -1].values

model = LogisticRegression(max_iter=1000)
model.fit(X, y)

odds = np.exp(model.coef_)
print("Odds:", odds)

points = np.array([[1.335, 0, 1, 0, 0, 109], [1.25, 0, 0, 1, 0, 279]])
probs = model.predict_proba(points)

for p, pr in zip(points, probs):
    print("Data point:", p.tolist(), "Probabilities:", pr.tolist())
```

CHAT TERMINAL

```
benwilliams@Bens-MacBook-Air-2 Exercise5 % /Users/benwilliams/Documents/Development/MachineLearning/.venv/bin/python /Users/benwilliams/Documents/Development/MachineLearning/HW/HW4/WilliamsBenjamin_HW4/Exercise5/exercise5.py
Final Coefficients: [-0.26023958  0.05335903  0.18666974]
Intercept: 189.89442410629354
benwilliams@Bens-MacBook-Air-2 Exercise5 % cd ../../..
benwilliams@Bens-MacBook-Air-2 HW % cd ..
benwilliams@Bens-MacBook-Air-2 MachineLearning % cd Lab 5
cd: string not in pwd: Lab
benwilliams@Bens-MacBook-Air-2 MachineLearning % ls
HW      Lab 1   Lab 2   Lab 5
benwilliams@Bens-MacBook-Air-2 MachineLearning % cd Lab\ 5
benwilliams@Bens-MacBook-Air-2 Lab 5 % cd WilliamsBenjamin_Lab5
benwilliams@Bens-MacBook-Air-2 WilliamsBenjamin_Lab5 % ls
Exercise1       Exercise2       Exercise3
benwilliams@Bens-MacBook-Air-2 WilliamsBenjamin_Lab5 % cd Exercise1
benwilliams@Bens-MacBook-Air-2 Exercise1 % /Users/benwilliams/Documents/Development/MachineLearning/.venv/bin/python "/Users/benwilliams/Documents/Development/MachineLearning/Lab 5/WilliamsBenjamin_Lab5/Exercise1/exercise1.py"
Odds: [[12.49500887  0.42784772]]
Data point: [7, 28] Probabilities: [0.24645934369530575, 0.7535406563046942]
Data point: [10, 34] Probabilities: [0.026606156436529083, 0.9733938435634709]
Data point: [2, 39] Probabilities: [0.9999999991172047, 8.827953286298422e-10]
benwilliams@Bens-MacBook-Air-2 Exercise1 % /Users/benwilliams/Documents/Development/MachineLearning/.venv/bin/python "/Users/benwilliams/Documents/Development/MachineLearning/Lab 5/WilliamsBenjamin_Lab5/Exercise1/exercise1.py"
Odds: [[12.49500887  0.42784772]]
Data point: [7, 28] Probabilities: [0.24645934369530575, 0.7535406563046942]
Data point: [10, 34] Probabilities: [0.026606156436529083, 0.9733938435634709]
Data point: [2, 39] Probabilities: [0.9999999991172047, 8.827953286298422e-10]
benwilliams@Bens-MacBook-Air-2 Exercise1 % cd ../Exercise2
benwilliams@Bens-MacBook-Air-2 Exercise2 % /Users/benwilliams/Documents/Development/MachineLearning/.venv/bin/python "/Users/benwilliams/Documents/Development/MachineLearning/Lab 5/WilliamsBenjamin_Lab5/Exercise2/exercise2.py"
Enter test size (e.g., 0.2): .2
Accuracy: 0.975
Labels order: [0, 1]
Confusion Matrix:
 [[ 91   5]
 [  0 104]]
benwilliams@Bens-MacBook-Air-2 Exercise2 % cd ../Exercise3
benwilliams@Bens-MacBook-Air-2 Exercise3 % ls
Bank-data.csv   exercise3.py
benwilliams@Bens-MacBook-Air-2 Exercise3 % /Users/benwilliams/Documents/Development/MachineLearning/.venv/bin/python "/Users/benwilliams/Documents/Development/MachineLearning/Lab 5/WilliamsBenjamin_Lab5/Exercise3/exercise3.py"
Odds: [[0.46240684 3.82416631 0.19212377 1.27540983 2.61213683 1.00682786]]
Data point: [1.335, 0.0, 1.0, 0.0, 0.0, 109.0] Probabilities: [0.8850920727445147, 0.11490792725548528]
Data point: [1.25, 0.0, 0.0, 1.0, 0.0, 279.0] Probabilities: [0.2547048300238929, 0.7452951699761071]
benwilliams@Bens-MacBook-Air-2 Exercise3 %
```

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